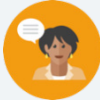


15.3 Probability

Lesson Introduction

 Benchmark	MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.		 Learning Targets	- I can express the probability of a chance event using a fraction, percentage, or decimal. - I can compare probabilities of chance events.	
 Benchmark Coherence	Building On			Working Toward	
	N/A			N/A	
 Essential Questions	- How do you find probability of an event? - How do you compare probabilities in different forms?				
 Lesson Narrative	In this lesson, students express the probability of events as fractions, decimals, and percentages. Students start with an Exploration on how to represent probability as a fraction, decimal, or percentage, relating these values ways of describing the likelihood of the event.				
 Component Summary	15.3.1 Warm-Up (~5 min)	Notice and Wonder about a set of cards (SE pg. 411)	15.3.4 Guided Instruction (10 min)	Expressing probability with fractions, decimals, and percentages (SE pg. 414)	
	15.3.2 Exploration (10 min)	Explore probability in different forms (SE pg. 412)	15.3.5 Try It (10 min)	Practice writing and comparing probabilities (SE pg. 415)	
	15.3.3 Bring It Together (5 min)	Summarize takeaways from Exploration (SE pg. 413)	15.3.6 Wrap-Up (~5 min)	Self-reflect on learning from the lesson (SE pg. 416)	
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Probability <u>Additional Terms:</u> - Sample space - Chance		N/A		N/A

 Teacher Tips	Common Misconceptions • Students may need support with converting between fractions, percentages, and decimals. • Students may struggle with the notation $P(\text{event})$ or mistake it as multiplication.	Things to Consider • Consider ways to make connections to describing the likelihood of events based on their probability throughout the remainder of the lesson after this is addressed in the 15.3.2 Exploration and 15.3.3 Bring It Together. • Model how to read probability notation to ensure students understand. • Students may need reminders about what vowels and consonants are.
	Literacy and Language Routines Think-Pair-Share In 15.3.5 Try It, consider implementing a Think-Pair-Share within question 1. This question contains a common misconception that students make when comparing probabilities. It is important to highlight this mistake. Students can pair up with a partner and share their thoughts before coming together as a class to discuss.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.2.1: Multiple Representations Throughout this lesson, students learn to express probabilities in a variety of ways. They connect this variety of representations to descriptions of the likelihood of an event, building foundational understanding for the remainder of the unit.
	 Differentiate Instruction	<i>When first introduced, the format $P(\text{event})$ may be confusing to students. Consider reading these statements aloud in the 15.3.4 Guided Instruction to help students understand what probability they are finding. i.e., $P(\text{vowel})$ – “What is the probability that a vowel will occur when Harper draws a letter out of the bag?”</i>
Lesson Notes:		

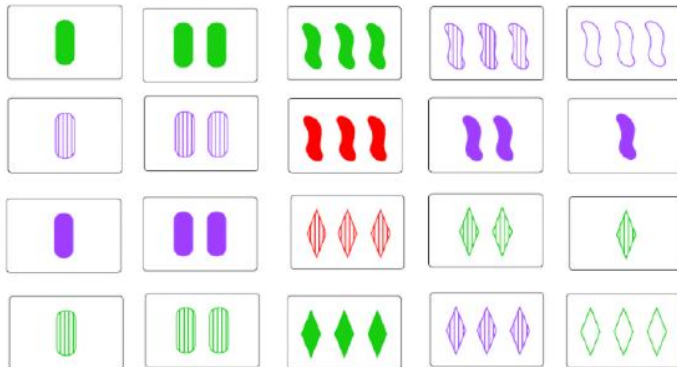
15.3.1 Warm-Up: Notice and Wonder

Component Overview: Students engage in a Notice and Wonder activity to activate the lesson.



15.3.1 Warm-Up: Notice and Wonder

1. Some of the cards in the card game SET are shown.



- a. What do you notice?

Answers will vary. I notice that the first three columns of cards have a pattern of increasing shapes on the cards by 1. I notice that there are less red than there are green and purple shapes.

- b. What do you wonder?

Answers will vary. I wonder if there is a pattern with the rest of the cards in the card game.



Notice and Wonder

Consider having students share out what they noticed with the class.



- If you were to select a card from those shown without looking, which number of shapes is the card most likely to have on it?

15.3.2 Exploration: Representing Probabilities

Component Overview: Students explore probability of events represented as a percentage, decimal, and fraction. Students are introduced to probability as it relates to the likelihood of an event happening and then explore what it means with a partner.



15.3.2 Exploration: Representing Probabilities

When performing simple experiments, all of the outcomes have the same likelihood of happening. For example, when flipping a fair coin, the sample space includes two possible outcomes, {heads, tails}. Each outcome has an equal chance of happening. Therefore, the chance of landing on heads is $\frac{1}{2}$ because this event represents 1 outcome out of 2 possible outcomes. Represented as a decimal or a percentage, the chance of landing on heads is 0.5 or 50%, respectively.

Work with your partner to complete the following.

- From the percentages listed below, determine the percent chance of each event occurring. Each value is used once.

0% 16. $\bar{6}$ % 33. $\bar{3}$ % 50% 100%

Event	Percent Chance
a. When given three doors with only one containing a prize behind it, what is the chance of opening the prize-winning door?	33. $\bar{3}$ %
b. When reaching into a dark closet with five pairs of shoes, what is the chance of pulling out a left shoe?	50%
c. When prompted to randomly select a crayon from a group of red, blue, and yellow crayons, what is the chance you choose a purple crayon?	0%
d. When rolling a fair six-sided die, what is the chance of landing on 1?	16. $\bar{6}$ %
e. When flipping a coin, what is the chance of landing on either heads or tails?	100%



Students may be tempted to write these as fractions instead of percentages.

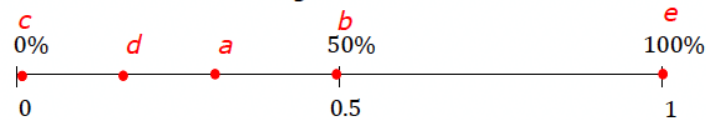
Students can think about the values in fraction form first, and then convert that fraction to the closest percent.



Check for student understanding that the 3rd scenario is 0% and that the 5th scenario is 100%. These are important ideas for students to grasp.

15.3.2 Exploration: Representing Probabilities (continued)

2. Plot and label a point to represent the percent chance of each event happening from the previous question. Use the letters labeling each event.



The **probability** of an event is a measure of the likelihood the event will occur. Probabilities are expressed using numbers from 0 to 1 as a **decimal, percentage, or fraction**.

3. Consider the event from question 1 that is least likely to occur.
- Which event is the least likely to happen?
Event c - choosing a purple crayon from a group of red, blue, and yellow crayons.
 - Describe, using sentences, the likelihood of this event happening.
Answers will vary. The likelihood of this event happening is impossible because there are 0 purple crayons to choose.
 - Discuss with your partner whether or not it is possible to have a probability less than this.
Answers will vary. It is not possible to have a probability less than this because impossible is the least likely an event can be.



- What does it mean for the probability to be 0?
- What is the range of possible probabilities?



*This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine students' thought processes to highlight or misconceptions to address.*

15.3.2 Exploration: Representing Probabilities (continued)

4. Consider the event from question 1 that is most likely to occur.

- a. Which event is the most likely to occur?

Event e - flipping a coin and it landing on either heads or tails.



- What does it mean to have a probability of 1?

- b. Describe, in sentences, the likelihood of this event happening.

Answers will vary. The likelihood of this event happening is certain because these are the only two possible outcomes. One or the other will always happen.



*This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine students' thought processes to highlight or misconceptions to address.*

- c. Discuss with your partner whether or not it is possible to have a probability greater than this.

Answers will vary. It is not possible to have a probability greater than this because 100% likelihood is the highest certainty.

5. Which event has the same likelihood of happening or not happening?

Event b - reaching into a dark closet with five pairs of shoes and pulling out a left shoe.

15.3.3 Bring It Together: Representing Probabilities

Component Overview: Students summarize the concepts they explored in the previous component to draw conclusions about the probability scale and what it looks like in the form of a decimal, fraction, or percentage.

15.3.3 Bring It Together: Representing Probabilities

1. Complete the statements.

The probability of an event can be any number between

0 and 1. A probability of 0 means it is ☐ certain ☐ impossible

for the event to occur. A probability of 1 means it is

☐ certain ☐ impossible the event will occur. The closer a

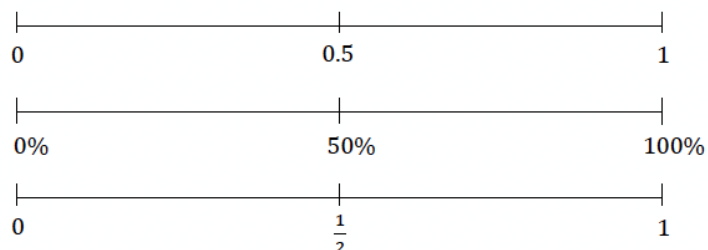
probability is to 1, the ☐ more ☐ less likely it is to occur,

while the closer it is to 0, the ☐ more ☐ less likely it is to

occur. A probability of 0.5 means the event is

☐ likely.
☐ unlikely.
☐ neither likely nor unlikely.

Probabilities can be represented by decimals, percentages, or fractions with equivalent values between 0 and 1.



Students should be able to work with their partner to complete question 1. The discussions in the previous component prime them for this summary. Review the number line together to ensure all students take away the necessary conclusions.



To connect from question 1, write “certain” above 1, “impossible” above 0, and “neither likely nor unlikely” above 0.5 in the space above the number lines when concluding this component with students.

15.3.4 Guided Instruction: Expressing Probabilities of Events

Component Overview: Students learn how to express the probability of an event, including the notation used. Students practice finding the probability of an event and converting the probability into different equivalent forms.



15.3.4 Guided Instruction: Expressing Probabilities of Events



The **probability** of an event occurring, notated as $P(\text{event})$, is determined by dividing the number of outcomes in the event by the possible outcomes in the sample space.

$$P(\text{event}) = \frac{\text{number of outcomes in the event}}{\text{total possible outcomes}}$$

Harper is playing a game where she draws a lettered tile out of a bag without looking.

The notation $P(A)$ in this context means “the probability of pulling out a tile with the letter A on it.”



- Find the probability of each event, represented as a fraction.

a. $P(A)$

$$\frac{1}{11}$$

b. $P(B)$

$$\frac{2}{11}$$

c. $P(M)$

$$\frac{0}{11}$$

d. $P(\text{vowel})$

$$\frac{5}{11}$$

e. $P(\text{letter})$

$$\frac{11}{11}$$



To extend students' thinking, consider having students draw a bag with the letters of their name in it.

- What would be the probability of choosing each letter?
- Would any letters have a higher probability than others?



This format $P(A)$ may be confusing to students at first. Consider reading it aloud to help students understand what the question is asking them to do.

I.e., $P(\text{vowel})$ – “What is the probability that a vowel will occur when Harper draws a letter out of the bag?”



The probability of pulling a vowel or consonant is purposefully included in this guided component. These terms are often used in simple probability events, so this provides an opportunity to ensure students know the meaning of these words from ELA. Note that “y” is considered a vowel in this word.

15.3.4 Guided Instruction: Expressing Probabilities of Events (continued)

2. Consider the probability of pulling a tile with a consonant on it.
 - a. Complete the table to find the fraction, decimal, and percentage equivalents of the probability.

Event	Fraction	Decimal	Percent
$P(\text{consonant})$	$\frac{6}{11}$	$0.\overline{54}$	$54.\overline{54}\%$

The probability of this event is a repeating decimal. When this occurs, the decimal or percentage representation is typically rounded.

- b. What is $P(\text{consonant})$ represented as a decimal rounded to the nearest hundredth?

0.55

- c. What is $P(\text{consonant})$ represented as a percentage rounded to the nearest tenth?

54.5%



Prompt students to use the terms “likely” or “unlikely” when referring to the probabilities.

Consider asking the following to solidify understanding:

- Which event from questions 1 or 2a is certain? Impossible? Unlikely? Somewhat likely?

15.3.5 Try It: Expressing Probabilities of Events in Different Forms

Component Overview: Students practice with expressing probabilities of events in different forms. Students also practice comparing probabilities in different forms, including fractions, decimals, and percentages.



15.3.5 Try It: Expressing Probabilities of Events in Different Forms

- The probability of rolling a 6 on a standard six-sided die is 16.67%, rounded to the nearest hundredth. Maisie says that this probability is greater than an event whose probability is 0.6 because $16.67 > 0.6$.

Explain whether you agree or disagree with Maisie's conclusion. Support your argument by showing your work.

Answers will vary. I disagree with Maisie because 16.67% as an equivalent decimal is $16.67\% \div 100 = 0.1667$, which is less than 0.6.



Think-Pair-Share

Question 1 contains a common misconception that students make when comparing probabilities in different forms. Consider implementing a Think-Pair-Share within question 1. Students can pair up with a partner and share their thoughts before coming together as a class to discuss if needed.



- How can you compare probabilities in different forms?

- The probabilities of five events are given.

A	B	C	D	E
40%	0.25	$\frac{5}{6}$	42.9%	$\frac{1}{5}$

- Order the values of the probabilities from least to greatest.

$\frac{1}{5}$, 0.25, 40%, 42.9%, $\frac{5}{6}$



Consider prompting students who are stuck to convert all probabilities to the same form so that they are easier to compare.

15.3.5 Try It: Expressing Probabilities of Events in Different Forms (continued)

- b. Match the probabilities with the events described in the table.

Event	Probability
 Landing on green on the spinner shown	25%
Pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles	$\frac{1}{5}$
Landing on a factor of 10 when rolling a fair ten-sided die	40%
Picking a card with a vowel on it from a set of cards with each letter in the word FLORIDA on it	42.9%
Landing on a number less than 6 on a standard six-sided die	$\frac{5}{6}$



Notice which students are having a hard time comparing probabilities. Consider using this information for differentiation strategies or groupings in the next lesson, as students will be using this skill.

- c. Explain which event is the least likely to occur based on the probabilities.

Answers will vary. The event that is least likely to occur is pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles because it has the smallest probability.

15.3.6 Wrap-Up: Reflection

Component Overview: Students reflect on their current level of understanding based on this lesson.



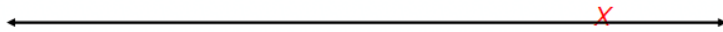
15.3.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can express the probability of a chance event using a fraction, percentage, or decimal.

Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
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2. I can compare probabilities of chance events.

Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
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The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.